

Name _____ Surname _____ Date ____/____/____

Energy and work · Exercise sheet

Use $g = 10 \text{ N/kg}$. Show every step of your working and write the unit in each answer. Give answers to 3 significant figures where needed.

1 · Kinetic energy

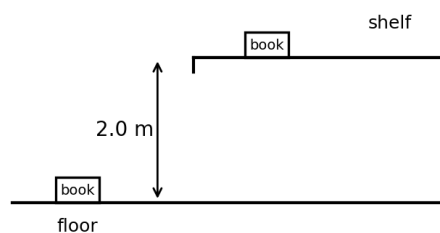
[4]

A cyclist and her bicycle have a total mass of 80 kg. She travels at a constant speed of 36 km/h.

- (a) Convert 36 km/h into m/s. [1]
- (b) Calculate the kinetic energy of the cyclist and bicycle. [2]
- (c) She speeds up until she is travelling twice as fast. By what factor does her kinetic energy increase? [1]

2 · Gravitational potential energy

[3]



A book of mass 1.5 kg is lifted from the floor onto a shelf 2.0 m above the floor.

- (a) Calculate the change in gravitational potential energy of the book. [2]
- (b) The book is then pushed 3.0 m along the shelf. What is the change in its gravitational potential energy during this push? Explain. [1]

3 · Speed and kinetic energy

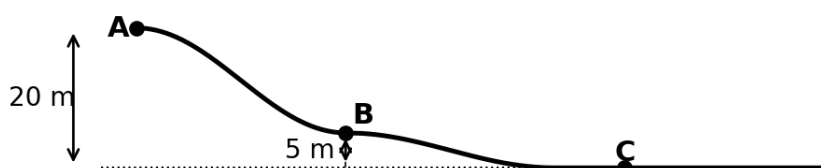
[4]

A car of mass 1200 kg covers a distance of 90 km at a constant speed of 72 km/h.

- (a) Calculate the time taken for the journey, in minutes. [2]
 - (b) Calculate the kinetic energy of the car. [2]
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4 · Energy changes on a roller coaster

[5]

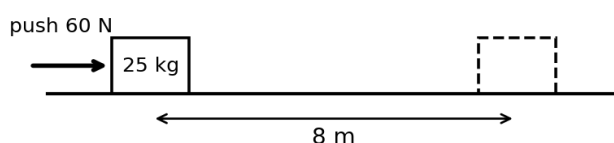


A roller-coaster cart of mass 400 kg starts from rest at point A, 20 m above the ground. Point B is 5 m above the ground and point C is at ground level. Ignore friction and air resistance.

- (a) Calculate the gravitational potential energy lost by the cart between A and B. [1]
- (b) State the kinetic energy of the cart at B and calculate its speed at B. [2]
- (c) Calculate the speed of the cart at C. [2]

5 · Work done

[4]

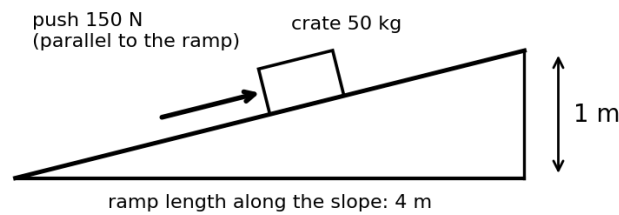


A student pushes a box of mass 25 kg along a horizontal floor with a constant horizontal force of 60 N. The box moves 8.0 m at constant speed.

- (a) Calculate the work done by the student. [2]
- (b) State the size of the friction force acting on the box. Explain your answer. [1]
- (c) The box does not gain kinetic energy. Where does the energy transferred by the student go? [1]

6 · The ramp

[4]

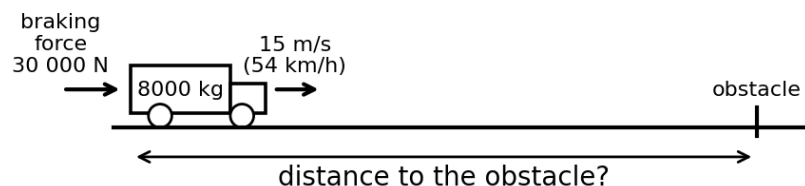


A worker pushes a crate of mass 50 kg up a ramp 4.0 m long, with a force of 150 N parallel to the slope. The crate is raised by 1.0 m.

- (a) Calculate the work done by the worker. [1]
- (b) Calculate the gain in gravitational potential energy of the crate. [1]
- (c) Calculate how much of the work done by the worker is not stored as potential energy, and suggest what happens to it. [2]

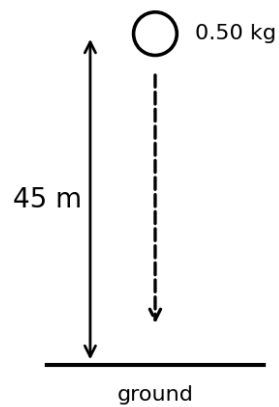
7 · Stopping a truck

[5]



A truck of mass 8000 kg travels at 54 km/h. The driver brakes. The braking force is constant and equal to 30 000 N.

- (a) Calculate the kinetic energy of the truck before braking. [2]
- (b) All this energy is removed by the braking force. Calculate the distance the truck travels while stopping. [2]
- (c) The truck now travels at twice the speed. Without a full calculation, state the stopping distance. Explain. [1]



A ball of mass 0.50 kg is dropped from rest from a height of 45 m. Ignore air resistance.

- (a) Calculate the loss of gravitational potential energy as the ball falls to the ground. [1]
- (b) State the kinetic energy of the ball just before it hits the ground. [1]
- (c) Calculate the speed of the ball just before it hits the ground, in m/s and in km/h. [2]

Total: 33 marks